The Maximum Number of Banks in a Four-Nation, Five-Currency Architecture

A Dimensional Theory of Banking, Counterfeiting, and Discount-Factor Networks

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Abstract

This paper develops a dimensional theory of banking within the previously established Four-Nation Bound and Five-Currency Ceiling frameworks.

The analysis begins from two prior results.

First, the degree-eight recoverable-money framework implies a maximum of four sovereign nations:

$$N_{\max} = 4.$$

Second, the military wage seigniorage framework implies a maximum of five viable reserve currencies:

$$C_{\max} = 5.$$

A bank is represented as a binary vector indicating which reserve currencies are present in positive quantities. The resulting banking state space is shown to contain

$$2^{C_{\max}}$$

elements.

For the five-currency architecture this yields

$$B_{\text{types}} = 32.$$

The introduction of nation-specific counterfeit replication generates a larger state space:

$$B_{\text{counterfeit}} = N_{\max} 2^{C_{\max}} = 128.$$

The resulting architecture admits interpretations in economics, finance, information theory, computer science, artificial intelligence, political science, geopolitics, warfare economics, network theory, philosophy, and the sciences.

The paper ends with “The End”

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1 Introduction

Two previous results establish structural upper bounds on global economic architecture.

The first result establishes:

$$N_{\max} = 4,$$

the maximum number of sovereign informational entities compatible with a degree-eight recoverable-money system.

The second result establishes:

$$C_{\max} = 5,$$

the maximum number of viable reserve currencies under finite reserve-rent constraints.

The present paper studies the next layer of organization:

$$\text{Nations} \to \text{Currencies} \to \text{Banks}.$$

The central question is:

How many distinct banks can exist within a four-nation, five-currency architecture?

2 Foundational Architecture

The previous papers imply

$$N_{\max} = 4$$

and

$$C_{\max} = 5.$$

Consequently the global reserve-currency set is

$$\mathcal{C} = \{c_1, c_2, c_3, c_4, c_5\}.$$

Each reserve currency possesses a discount factor

$$D_i(t), \qquad i = 1, \dots, 5.$$

The discount factors represent the temporal valuation structure of the global monetary system.

3 Bank Representation

3.1 Binary Representation

Define

$$b_i = \begin{cases} 1, & \text{currency } i \text{ present,} \\ 0, & \text{currency } i \text{ absent.} \end{cases}$$

A bank is represented by

$$b = (b_1, b_2, b_3, b_4, b_5).$$

Therefore

$$b \in \{0, 1\}^5.$$

3.2 Discount-Factor Representation

The complete banking state is

$$B(t) = (b_1 D_1(t), b_2 D_2(t), b_3 D_3(t), b_4 D_4(t), b_5 D_5(t)).$$

Notice that quantities of currencies are unnecessary.
Only the presence or absence of each discount-factor process matters.

4 The Banking State Space

The set of all banks is

$$\mathcal{B} = \{0, 1\}^5.$$

The cardinality is

$$|\mathcal{B}| = 2^5.$$

Hence

$$B_{\text{types}} = 32.$$

Theorem 1 (Banking State-Space Theorem). *Let $C_{\max}$ denote the maximum number of currencies. Then the number of distinct bank types is*

$$B_{types} = 2^{C_{\max}}.$$

Proof. Each currency may be either present or absent.
Therefore each coordinate has two possible states.
Since there are $C_{\max}$ coordinates,

$$|\mathcal{B}| = 2^{C_{\max}}.$$

The result follows immediately. $\square$

5 Five-Currency Banking Architecture

Substituting

$$C_{\max} = 5$$

yields

$$B_{\text{types}} = 2^5 = 32.$$

The 32 states correspond to all subsets of

$$\{c_1, c_2, c_3, c_4, c_5\}.$$

Examples include

$$00000,\ \ 10000,\ \ 11001,\ \ 11111.$$

6 Counterfeiting

Suppose nation j can create a counterfeit copy of any bank.
Let

$$j \in \{1, 2, 3, 4\}.$$

The state becomes

$$(j, b).$$

The counterfeit banking space is

$$\mathcal{N} \times \mathcal{B}.$$

Since

$$|\mathcal{N}| = 4$$

and

$$|\mathcal{B}| = 32,$$

we obtain

$$|\mathcal{N} \times \mathcal{B}| = 128.$$

Theorem 2 (Counterfeit Banking Theorem). *The maximum number of counterfeit bank states is*

$$B_{counterfeit} = N_{\max} 2^{C_{\max}}.$$

Proof. Each bank type can be replicated independently by every nation.

Therefore

$$B_{\text{counterfeit}} = |\mathcal{N}||\mathcal{B}|.$$

Substituting

$$|\mathcal{N}| = N_{\max}$$

and

$$|\mathcal{B}| = 2^{C_{\max}}$$

gives

$$B_{\text{counterfeit}} = N_{\max} 2^{C_{\max}}.$$

□

7 Fundamental Banking Counting Theorem

Combining the Four-Nation Bound and Five-Currency Ceiling:

$$N_{\max} = 4,$$

$$C_{\max} = 5.$$

Hence

$$B_{\text{types}} = 2^5 = 32,$$

and

$$B_{\text{counterfeit}} = 4 \times 2^5 = 128.$$

Theorem 3 (Fundamental Banking Counting Theorem)**.** *For a system with*

$$N_{\max}$$

nations and

$$C_{\max}$$

currencies,

$$B_{types} = 2^{C_{\max}}$$

and

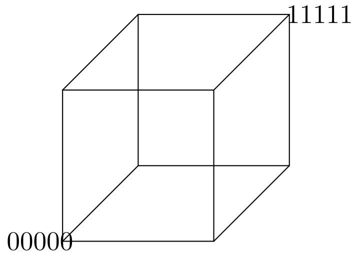
$$B_{counterfeit} = N_{\max} 2^{C_{\max}}.$$

8 Hypercube Geometry

The banking state space is

$$\{0,1\}^5.$$

This is precisely the vertex set of a five-dimensional hypercube.



The complete hypercube contains

$$2^5 = 32$$

vertices.

Counterfeiting produces four copies of the hypercube.

9 Economics

Banks become combinatorial monetary institutions.

Each additional reserve currency doubles the banking state space:

$$B_{\text{types}} = 2^{C_{\max}}.$$

Therefore reserve-currency proliferation generates exponential growth in banking complexity.

10 Finance

The discount-factor vector

$$(D_1(t), \ldots, D_5(t))$$

acts as the temporal backbone of the banking system.

Banks differ not through quantity holdings but through the subset of discount-factor processes they possess.

11 Physics

The banking system resembles a five-spin Ising system.

Each coordinate satisfies

$$b_i \in \{0, 1\}.$$

The banking state space therefore resembles a discrete energy landscape.

12 Chemistry

Currencies behave analogously to chemical species.

Banks correspond to molecular combinations.

Five species produce

$$2^5$$

possible compounds.

13 Biology

The binary currency vector resembles a genetic sequence.

Currencies act as genes.

Banks act as phenotypes.

14 Psychology and Psychiatry

Humans simplify complex decision environments through categorical representations.

The binary currency representation minimizes cognitive load while preserving essential structure.

15 Statistics

The banking architecture admits a Bernoulli representation:

$$b_i \sim \mathrm{Bernoulli}(p_i).$$

Expected currency count:

$$E[K] = \sum_{i=1}^{5} p_i.$$

16 Computer Science

The banking vector is a bit string.

Storage requirement:

$$5$$

bits per bank.

The complete state space contains

$$2^5$$

states.

17 Algorithms

17.1 Pseudo-Code

```
INPUT:
Currencies = 5
Nations = 4

BankTypes = 2^Currencies

CounterfeitBanks =
Nations * BankTypes

OUTPUT:
32
128
```

18 Data Science

Banks become categorical observations in a 32-state feature space.

The state vector is naturally suitable for clustering, classification, and anomaly detection.

19 Artificial Intelligence

The banking vector is equivalent to a binary embedding:

$$b \in \{0,1\}^5.$$

The counterfeit architecture becomes

$$(j, b).$$

This may be interpreted as a structured latent representation.

20 Political Science

Each counterfeit bank represents an institutional imitation mechanism.

Nations replicate monetary institutions developed elsewhere.

21 International Relations

The architecture implies

$$4 \times 32 = 128$$

possible nation-bank combinations.

This defines a finite geopolitical banking space.

22 Warfare Economics

Reserve currencies generate seigniorage.

Seigniorage supports military capability.

Banks become nodes through which monetary power propagates.

23 Network Theory

Define graph

$$G = (V, E).$$

Each bank is a vertex.

Therefore

$$|V| = 32$$

or

$$128$$

under counterfeiting.

24 Philosophy

The theory illustrates a broader principle:

> Finite informational structures generate finite institutional architectures.

The banking system emerges as a combinatorial consequence of currency scarcity and sovereign scarcity.

25 Comparative Table

Quantity	Formula	Value
Maximum Nations	$N_{\max}$	4
Maximum Currencies	$C_{\max}$	5
Bank Types	$2^{C_{\max}}$	32
Counterfeit Banks	$N_{\max}2^{C_{\max}}$	128

26 Conclusion

Starting from

$$N_{\max} = 4$$

and

$$C_{\max} = 5,$$

we established:

1. Banks are binary currency subsets.
2. Discount factors determine currency identity.
3. Quantities are unnecessary.
4. The banking state space contains
$$32$$
bank types.
5. Counterfeiting expands the architecture to
$$128$$
nation-bank states.

The resulting structure links economics, finance, information theory, computer science, artificial intelligence, warfare economics, and political science through a common counting framework.

The End.

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Glossary

$N_{\max}$ Maximum number of nations.

$C_{\max}$ Maximum number of reserve currencies.

$B_{\mathbf{types}}$ Number of bank types.

$B_{\mathbf{counterfeit}}$ Number of counterfeit banking states.

$D_i(t)$ Discount factor of currency i.

b_i Currency-presence indicator.

$\mathcal{B}$ Bank state space.

$\mathcal{N}$ Nation set.

G Banking network graph.

The End